

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?
- Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?
- Estimate the price elasticity of demand "at the means" using the results from (b).
- Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.
- What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?
- Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

(a)

Inverse Demand Equation:

$$Q = \alpha_1 + \alpha_2 P + \alpha_3 PS + \alpha_4 DI + e_d$$

$$\rightarrow \alpha_2 P = Q - \alpha_1 - \alpha_3 PS - \alpha_4 DI - e_d$$

$$\rightarrow P = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q - \frac{\alpha_3}{\alpha_2} PS - \frac{\alpha_4}{\alpha_2} DI - \frac{e_d}{\alpha_2}$$

$$\rightarrow P = a_1 + a_2 Q + a_3 PS + a_4 DI + e_d$$

$$a_2 = -\frac{\alpha_1}{\alpha_2} < 0 ; a_3 = \frac{1}{\alpha_2} > 0 ; a_4 = -\frac{\alpha_4}{\alpha_2} > 0$$

Inverse Supply Equation:

$$Q = \beta_1 + \beta_2 P + \beta_3 PF + e_s$$

$$\rightarrow \beta_2 P = Q - \beta_1 - \beta_3 PF - e_s$$

$$\rightarrow P = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} Q - \frac{\beta_3}{\beta_2} PF - \frac{1}{\beta_2} e_s$$

$$\rightarrow P = b_1 + b_2 Q + b_3 PF + e_s$$

$$b_2 = \frac{1}{\beta_2} > 0 ; b_3 = -\frac{\beta_3}{\beta_2} > 0$$

(b)

需求反函數和供給反函數的 **sign** 皆符合我們的預期

需求反函數在 **0.01** 的顯著水準下，只有 **ps** 和 **di** 的係數是顯著的，但 **q** 的係數並不顯著。

供給反函數所有係數在 **0.01** 的顯著水準下皆顯著。

```
> summary(demand_2s1s)

Call:
ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-39.661  -6.781   2.410   8.320  20.251

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -11.428     13.592  -0.841  0.40810
q             -2.671      1.175  -2.273  0.03154 *
ps              3.461      1.116   3.103  0.00458 **
di             13.390      2.747   4.875  4.68e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567,    Adjusted R-squared: 0.5056
Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06
```

```
> summary(supply_2s1s)

Call:
ivreg(formula = p ~ q + pf | ps + di + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-9.7983  -2.3440  -0.6281   2.4350  11.1600

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -58.7982     5.8592  -10.04 1.32e-10 ***
q              2.9367     0.2158   13.61 1.32e-13 ***
pf              2.9585     0.1560   18.97 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486,    Adjusted R-squared: 0.9448
Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16
```

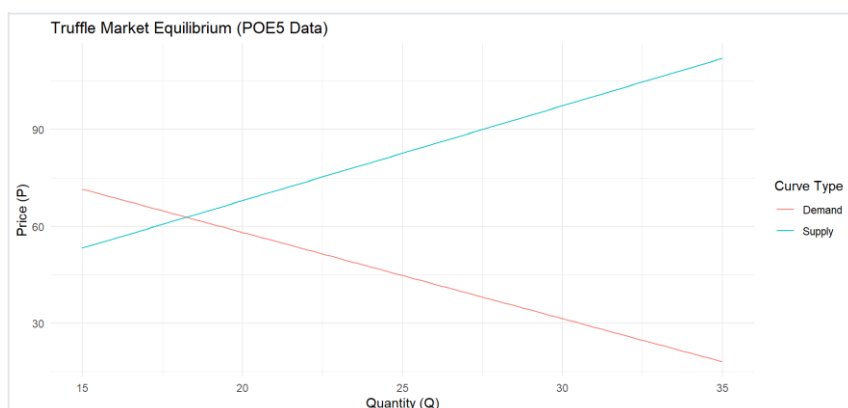
(c)

```
> # 3. (c) 計算平均值處的需求價格彈性 -----
> a2_hat <- coef(demand_2s1s)["q"]
> p_mean <- mean(truffles$p)
> q_mean <- mean(truffles$q)
>
> # 彈性 = (dq/dP) * (P/Q)
> elasticity_d <- (1 / a2_hat) * (p_mean / q_mean)
> cat("\n需求價格彈性 (at means):", round(elasticity_d, 4), "\n")
```

需求價格彈性 (at means): -1.2725

>

(d)



(e)

```
> # 方法 1：使用 (d) 小題的結構係數解聯立方程
> # 需求： $P = d\_int + d\_slope * Q$ 
> # 供給： $P = s\_int + s\_slope * Q$ 
> #  $d\_int + d\_slope * Q = s\_int + s\_slope * Q$ 
> # 解  $Q = (s\_int - d\_int) / (d\_slope - s\_slope)$ 
>
> d_slope <- coef(demand_2sls)["q"]
> s_slope <- coef(supply_2sls)["q"]
>
> Q_eq <- (s_int - d_int) / (d_slope - s_slope)
> P_eq <- d_int + d_slope * Q_eq
>
> cat("--- (e) 結構方程式得到的均衡值 ---\n")
--- (e) 結構方程式得到的均衡值 ---
> cat("均衡數量  $Q^*$  =", round(Q_eq, 4), "\n")
均衡數量  $Q^*$  = 18.2502
> cat("均衡價格  $P^*$  =", round(P_eq, 4), "\n")
均衡價格  $P^*$  = 62.8426
~

> # 方法 2：使用簡化式 (Reduced-form) 預測
> # 這通常是直接對  $Q$  和  $P$  分別做外生變數 ( $PS$ ,  $DI$ ,  $PF$ ) 的 OLS
> # 這裡示範如何用 predict 函數 (假設我們已經跑過簡化式模型)
>
> # 建立預測用的新資料點
> new_data <- data.frame(ps = 22, di = 3.5, pf = 23)
>
> # 估計簡化式模型 (Reduced Form)
> rf_q <- lm(q ~ ps + di + pf, data = truffles)
> rf_p <- lm(p ~ ps + di + pf, data = truffles)
>
> Q_rf <- predict(rf_q, newdata = new_data)
> P_rf <- predict(rf_p, newdata = new_data)
>
> cat("\n--- (e) 簡化式模型預測的均衡值 ---\n")

--- (e) 簡化式模型預測的均衡值 ---
> cat("預測  $Q^*$  =", round(Q_rf, 4), "\n")
預測  $Q^*$  = 18.2604
> cat("預測  $P^*$  =", round(P_rf, 4), "\n")
預測  $P^*$  = 62.8154
> |
```

結構式 (2SLS) 與簡化式所得到的均衡值高度吻合。這在理論上是合理的，因為 2SLS 估計量本質上就是基於簡化式的關係推導而來。這證明了模型預測的一致性。

(f)

需求反函數 **sign** 中，**q** 的係數不符合我們的預期。

而其他的係數皆符合預期。

需求反函數在 **0.01** 的顯著水準下，只有 **di** 的係數是顯著的，但 **q** 和 **ps** 的係數並不顯著。

供給反函數所有係數在 **0.01** 的顯著水準下皆顯著。

對比 (b) 與 (f) 的結果可以發現，**普通最小平方法 (OLS)** 無法有效估計需求函數，其 **q** 的係數符號不符合經濟直覺且統計上不顯著。這是因為價格與數量是市場均衡下同時決定的，存在嚴重的**內生性問題**。

相比之下，(b) 小題採用的 **2SLS** 方法，透過引入供給面的外生變數（如生產成本 **PF**）作為工具變數，成功修正了這種偏誤，使得需求曲線的斜率恢復為合理的負值。這說明了在處理連立方程模型時，使用工具變數法（如 **2SLS**）對於獲取一致且可靠的參數估計是至關重要的。

```
> summary(demand_ols)

Call:
lm(formula = p ~ q + ps + di, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-25.0753  -2.7742  -0.4097   4.7079  17.4979

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.6195     9.0872  -1.499   0.1460
q             0.1512     0.4988   0.303   0.7642
ps            1.3607     0.5940   2.291   0.0303 *
di           12.3582     1.8254   6.770 3.48e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared:  0.8013,    Adjusted R-squared:  0.7784
F-statistic: 34.95 on 3 and 26 DF,  p-value: 2.842e-09

> summary(supply_ols)

Call:
lm(formula = p ~ q + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-8.4721 -3.3287  0.1861  2.0785 10.7513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -52.8763     5.0238 -10.53 4.68e-11 ***
q             2.6613     0.1712  15.54 5.42e-15 ***
pf            2.9217     0.1482  19.71 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared:  0.9531,    Adjusted R-squared:  0.9496
F-statistic: 274.4 on 2 and 27 DF,  p-value: < 2.2e-16
```